

Tianjiao Wang

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EDUCATION

Wen-zhou kean University

September 1, 2022-Present

- Bachelor of Compute Science | **GPA:** 3.8 / 4.0 | **Rank:** 12 / 136 (Top 10%)
- Relevant Courses: Computer Programming, Data Structures, Operating Systems, Linux and Network Security, Algorithm Analysis, Big Data Technology, Calculus, Statistics, Linear Algebra, Data Analysis, Numerical Analysis

HONORS & AWARDS

- CET - 6 Certificate (2026)
- Third - Class Dean's Scholarship (Top 10% of cohort), Wenzhou-Kean University (2024 – 2025)
- Student Research Day Presenter Recognition, Wenzhou-Kean University (2025)
- Second Prize (Group Award), Instrumental Ensemble Category (Group A), Zhejiang Provincial College Student Art Festival (2023)
- CET - 4 Certificate (2023)

RESEARCH ACTIVITIES

Academic papers

Tianjiao Wang. (2026) *Robust and Calibrated Machine Learning Models for Alzheimer's Disease Risk Prediction.*

2026 7th International Conference on Big Data & Artificial Intelligence & Software Engineering

- Implemented a robustness-oriented machine learning framework on 2,149 clinical records to predict Alzheimer's disease risk.
- Improved probabilistic reliability compared to linear models; reduced Brier score by over 58% ($0.1192 \rightarrow 0.0501$) using XGBoost.

Tianjiao Wang. Ming Chen*. (2026) *Data-Driven Characterization of Urban Air Pollution Exposure and Bio-Health Pressure Patterns via Probabilistic Modeling.*

2026 2nd International Conference on Big Data and Intelligent Medicine (BDIMed 2026)

- Developed a two-stage probabilistic modeling framework using Gaussian Mixture Models (GMM) to identify latent multi-pollutant exposure regimes across 8 global megacities (88,489 observations), with strong cluster tendency validated by Hopkins statistic ($H = 0.865$) and variance inflation factors ≈ 1.0 indicating independence among AQI , $PM_{2.5}$, PM_{10} , NO_2 , and O_3 .
- Performed model selection via BIC/AIC and posterior stability analysis, selecting a six-component solution ($K = 6$) with mean maximum posterior probabilities ranging from 0.743–0.803; PCA revealed particulate matter dominating PC1 and oxidizing pollutants shaping PC2, confirming multimodal structural heterogeneity in pollutant composition.
- Identified a monotonic gradient between exposure intensity and healthcare pressure: high-intensity clusters ($n = 29,842; 19,558$) exhibited 26–27% high-pressure hospitalization rates, compared to 16–26% in low-intensity clusters; findings provide empirical support for differentiated emission-control strategies and health-resource allocation policies.

Capstone

Generating Structure-Consistent Hard Sample as Data Source for Medical Image Analysis December 2025-Present

- Proposed a unified diffusion-based hard-sample generation framework integrating ROI-guided local inpainting, Attention Discrepancy Maximization (ADM), prototype-level semantic shifting, and DADA-style adversarial perturbation to jointly enforce geometric alignment, semantic difficulty, and controllable false-positive synthesis.
- Designed a low-data experimental regime in which the diffusion model is trained on 50 real medical images and evaluated across 5 public polyp segmentation datasets (2,248 images total) to assess cross-model generalization under limited supervision.
- Aimed to improve downstream segmentation robustness (mIoU, Dice, F1) and reduce false-positive sensitivity under cross-dataset domain shift by generating anatomically consistent yet semantically challenging hard samples.

Team project

cfDNA Threshold Optimization in Non-Invasive Prenatal Testing

September 2025

- Analyzed large-scale NIPT clinical dataset to model cfDNA concentration dynamics; applied mixed-effects regression and survival analysis to quantify BMI dilution effects and gestational nonlinear growth trends ($p < 0.01$).
- Developed optimal testing time recommendations using Cox and AFT models with interval censoring; achieved $>90\%$ cfDNA attainment probability across BMI subgroups.
- Built an aneuploidy screening classifier with XGBoost integrating sequencing QC metrics; addressed class imbalance and improved detection robustness on skewed clinical samples.

COMMUNITY ENGAGEMENT

Teaching Assistant, Wenzhou-kean University, Wenzhou, China September 1, 2025 - Present

- Graded programming assignments and reviewed code submissions for CPS 3410 (Applied Algorithms & Data Structures) and CPS 4150 (Computer Architecture) for 100+ students.
- Assisted in exam proctoring and grading to maintain course evaluation quality.

Data Analyst, Beijing Qingpu Intelligent Technology, Beijing, China December 23, 2024 - January 25, 2025

- Extracted and standardized key data indicators from enterprise Excel spreadsheets; utilized Python web scraping techniques to batch extract industry research literature from scientific websites. Implemented ETL processes to accurately import data into the DBeaver database, enabling real-time synchronization with the front-end platform via backend API interfaces, thus enhancing automation and timeliness of data processing.
- Processed Excel data using Python's pandas library and optimized SQL queries to improve the specificity and clarity of query results. Improved the database documentation for the Digital Industry Information Platform by creating and updating data tables to ensure a clear and standardized database structure that supports data analysis and reporting.
- Used a data ingestion tool to import Excel data, performing data cleaning and mapping to the target database based on the correct format and corresponding indicators. Enhanced the mapping dictionary to improve data clarity and traceability.

Data Engineer, Shanghai Qijiang Information Technology, Shanghai, China June 1, 2024 - August 31, 2024

- Used MySQL as the primary data source, configured data connections and scheduled cron tasks to regularly migrate customer data from the testing environment to the operational data store (ODS) in the data center, ensuring data stability and supporting further analysis.
- Utilized SQL query tools and data model configurations to define data standards and build dimensional modeling tables, providing structured support to the detailed data layer (DWD) of the data warehouse, thereby improving data consistency and usability.
- Integrated multi-source information from the ODS layer via efficient ETL pipelines, performed data governance tasks to ensure accurate relationships between heterogeneous data sources, enhancing data quality and accessibility.

EXTRA CURRICULUM ACTIVITIES

Teaching Assistant, Google & Datawhale Campus AI Event April 2026

- Supported hands-on deployment sessions for 50+ participants at a joint campus AI event co-organized by GDG Wenzhou and Datawhale.
- Guided participants through cloud deployment of Google's Gemma open-source model, covering environment setup, inference configuration, and real-time troubleshooting.
- Hands-on with Datawhale's Open Claw AI Agent framework, practicing automated workflow design and enterprise platform integration (Feishu, DingTalk).

The 12th (2026) National College Student Statistical Modeling Competition (Awaiting Results) April 2026

Project Name: Text Mining and Quantitative Analysis on A-Share Analyst Reports

- Constructed a "Foreign Market Uncertainty Index" by applying text mining to 2005 – 2024 analyst reports, identifying vague language and risk sentiment, and extracting core uncertainty scores (Unct_score).
- Designed a Chinese financial text analysis model with NER and keyword extraction for granular risk quantification.
- Conducted regression analysis with firm-level data and fixed effects; results showed uncertainty reduces investment and profitability while weakly increasing R&D spending.

2026 Mathematical Contest in Modeling (MCM/ICM) S Award February 2026

Project Name: An Integrated Decision Framework for Professional Sports Teams

- Designed a Dynamic Multi-Layer Decision Model (Q-Model) integrating profit maximization and long-term franchise valuation under salary cap and market constraints; derived closed-form optimal investment policy and quantified valuation-leverage effects on optimal spending behavior.
- Developed a Player Acquisition Optimization Framework using PCA-based Player Value Index (13 performance metrics) and Monte Carlo simulation; results show free-agent strategies achieve expected roster value gain ≈ 230 units (budget slack $> \$20M$), while draft strategies maintain $CVaR10\% \approx 100-110$ and $ROI > 40\%$, yielding an optimal hybrid allocation of 20–30% budget to free agency.
- Implemented a Reinforcement Learning-based Dynamic Ticket Pricing Model, achieving \$413.67M total revenue over 41 games ($\uparrow 123.4\%$ vs. fixed pricing baseline) with 100% sell-out rate and stable performance under $\pm 10\%$ star-effect sensitivity.